Anqi Bao

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Summary

Machine learning researcher with 5+ years of industry experience building large-scale models in adversarial, safety-critical domains. Strong track record of designing and deploying deep learning systems using transformers, graph neural networks, and representation learning for fraud detection, anomaly detection, and time-series modeling. Cross-disciplinary training in engineering, statistics, and computer science; published first-author work on graph embeddings and deep sequence models. Comfortable owning projects end-to-end: framing problems, implementing models, running experiments, and communicating results. Eligible to work in the US (Green Card holder).

Experience

The Trade Desk — ***Senior Applied Scientist***
Nov 2025 – Present

- Conduct research on and develop large-scale identity graphs (that powers downstream real-time bidding and ranking), working with billions of nodes and edges to detect anomalies, evaluate stability, and analyze emergent structure

Microsoft — ***Senior Applied Scientist***
Jun 2021 – Oct 2025

- Fine-tuned LLaMA-2 using parameter-efficient techniques (LoRA) to learn a proprietary fraud query language for automatic rule generation, enabling customers to define custom fraud-detection rules with natural language inputs.
- Engineered a pipeline to convert natural language-based features into dense embeddings using pre-trained LLMs (e.g., BERT) for fraud detection. Enhanced the pipeline with techniques such as dimensionality reduction to retain key signal components and reduce noise, resulting in measurable improvements in model accuracy and robustness
- Research and apply attention-based (Transformer) technique to extract hidden time series information from day-to-day transactional data and integrated in model to find anomaly patterns
- Designed graph neural network pipelines (DeepWalk, GAT) for large-scale entity linkage and anomaly discovery, improving detection of coordinated malicious actors. Work published in Microsoft Journal of Applied Research.
- Researched and developed a bot detection model leveraging anomaly patterns in device fingerprinting and mouse behavior logs to identify Microsoft accounts potentially created by bots, enhancing automated fraud prevention capabilities
- Develop customized data/modeling/visualization Azure pipelines for customers with new fraud scenarios and standardize the procedure for model training and impact analysis

Amazon — ***Applied Scientist***
Feb 2021 – Jun 2021

- Research, design and develop innovative production grade Machine Learning models on AWS to prevent denied parties from transacting on Amazon
- Understand the business requirements, translate them into complex analytical outputs and esign tests to explain performance of models quantitatively
- Working in a highly collaborative environment partnering with various science, product management, engineering, operations, finance, business intelligence and analytics teams to develop ML systems
- Building, testing and improving prototypes given the feedback from QA and internal investigators

Microsoft — ***Data Scientist***
Jul 2020 - Feb 2021

- Involved with entire pipeline of offline proof of concept fraud protection for potential customers. Hands on experience with Databricks (Pyspark) and Azure Pipeline for data cleaning, data mapping, feature calculation, modeling building and results visualization
- Regular Feature validation (with Cosmos DB and Scope) and model score monitor (with Power BI) for real time fraud protection for major customers

Halliburton Landmark R&D — ***Software Development Engineer***

Aug 2019 - May 2020

- Web/cloud based UI development with Node JS and Angular JS for commercial reservoir simulator
- Couple functions in C++/JAVA/JavaScript into RESTful API backend framework
- Well control optimization with reinforcement learning methods

Halliburton Landmark ***Software Engineer Intern***
Jun 2018 - Aug 2018

- Computer Vision and Deep Learning Application on Wellbore Schematics using PyTorch
- Develop prototype for extracting objects information from schematics and convert it to structured data

Education

Georgia Institute of Technology *M.S. in Computer Science*
2021 – 2025
Texas A&M University *Ph.D. in Petroleum Engineering*
2015 – 2019 GPA: 4.0 / 4.0
Texas A&M University *M.S. in Statistics*
2017 – 2019 GPA: 3.8 / 4.0

Skills

Languages: Python, C++, R, SQL
Machine Learning: PyTorch, TensorFlow, Scikit-learn
Tools/Infra: Azure, AWS, Hadoop, Spark, Kafka
Specialties: Machine Learning, Deep Learning, LLM, Reinforcement Learning, Agentic AI

Selected Publications

- 2022, Enhancing Fraud Detection Using Graph Embeddings. **First Author**, Microsoft Journal of Applied Research, Vol 16 Article 95.
  Summary: DeepWalk and Graph Neural Networks (more specifically Graph Attention Network) as graph embedding techniques for generating node embeddings for real-time and batch mode evaluations. Our results demonstrate the ability for these techniques to capture essential linkage information in transactional data and provide the possibility of using the techniques in fraud protection settings.
- 2020, Data-driven End-to-end Production Prediction of Oil Reservoirs by EnKF-enhanced Recurrent Neural Networks, **First Author**, Society of Petroleum Engineering LACPEC conference
  Summary: Combine Deep learning with Ensemble Kalman Filter (EnKF) to develop real-time proxy model for highly nonlinear dynamic system. An RNN model is successfully trained using time series production data. The neural network is optimized with Bayesian optimization and enhanced by EnKF for robustness.
- 2019, Wellbore Schematics to Structured Data Using Artificial Intelligence Tools. **Second Author**, Offshore Technology Conference
  Summary: Use the RetinaNet model for object detection and combine with character detection tools to fully interpret and regenerate wellbore schematic. This product significantly reduces manual effort and human errors.

Projects

- **Distributed Trade System**: Built fully functional online trade system with SpringBoot, MyBatis, Elasticsearch, RabbitMQ, Redis, AWS.
- **Wellbore Object Detection**: Used RetinaNet and character detection tools to interpret and regenerate schematics, reducing manual effort and errors.
- **Reservoir Simulator UI**: Developed Angular JS front end and Express API backend for Halliburton's simulator; collaborated globally using Agile practices.
- **Predictive Modeling**: Combined Deep learning with Ensemble Kalman Filter (EnKF) to develop real-time proxy model for nonlinear dynamic systems; optimized with Bayesian methods.

Google Scholar

https://scholar.google.com/citations?user=8oSmAk4AAAAJ&hl=en